

1. Describe the working of a bridge rectifier with and without capacitor filter, with a suitable diagram. Definition: A bridge rectifier is a full-wave rectifier circuit that converts alternating current (AC) into direct current (DC) using four diodes arranged in a bridge configuration, and it does not require a centre-tapped transformer, unlike a two-diode full-wave rectifier. Circuit diagram description: The AC supply is fed through a step-down transformer to points A and B of a diamond-shaped diode bridge consisting of D1, D2, D3, and D4. The load resistor RL is connected across the other diagonal of the bridge, between the junction of D1–D4 and the junction of D2–D3. Working during positive half cycle: Point A becomes positive with respect to point B. This forward-biases diodes D1 and D3, allowing them to conduct current through the load, while D2 and D4 remain reverse-biased and block current flow. Working during negative half cycle: Point B becomes positive with respect to point A. This forward-biases diodes D2 and D4, which conduct current through the load in the same direction as before, while D1 and D3 are reverse-biased and blocked. Nature of output: Because the load current flows in the same direction during both half cycles, the output across RL is always unidirectional pulsating DC, and the ripple frequency of the output is twice the supply frequency, that is, 100 Hz for a 50 Hz AC input. Working without capacitor filter: The output waveform consists of a series of half-sine pulses that rise from zero to peak and fall back to zero every half cycle, giving DC with very high ripple content, which is unsuitable for powering sensitive electronic circuits that require a steady voltage. Working with capacitor filter: A capacitor connected in parallel with the load charges up to the peak value of the rectified voltage during conduction and discharges slowly through the load resistance when the diodes stop conducting, filling in the gap between successive pulses. This smooths the pulsating DC into a nearly steady output voltage with only a small ripple, and using a larger capacitance value gives a lower ripple and a smoother, more constant DC output. Advantages: The bridge rectifier does not need a centre-tapped transformer, gives better transformer utilisation, and has higher efficiency compared to half-wave rectifiers, which makes it the most widely used rectifier circuit in DC power supplies.	2. Compare the features and applications of DOL starter and Star-Delta starter. DOL starter construction and working: A Direct-On-Line starter connects the motor directly across the full three-phase supply voltage through a contactor and an overload relay. When the start button is pressed, the contactor closes instantly and applies full line voltage to the stator windings without any voltage reduction. DOL starter features: The starting current drawn is very high, typically 6 to 8 times the full-load current, along with a high starting torque of about 4 to 6 times the full-load torque. The starter is simple in construction, low in cost, and easy to maintain, since it consists mainly of a contactor, an overload relay, and start/stop push buttons. DOL starter limitations and application: The heavy inrush current causes a significant voltage dip on the supply network and produces a sudden mechanical jerk on the driven machinery, so DOL starters are recommended only for small motors, generally up to 5 HP, where the supply system can tolerate the starting surge. Star-Delta starter construction and working: A Star-Delta starter first connects the stator windings in star configuration during starting, which reduces the voltage across each winding to about 58 percent (1/√3) of the rated line voltage. After a preset time delay controlled by a timer relay, or once the motor reaches near-rated speed, the connection automatically switches to delta, applying full line voltage for normal running. Star-Delta starter features: Because of the reduced starting voltage, the starting current and starting torque are both reduced to roughly one-third of the values that would occur with a DOL start, giving a smoother start with less voltage dip and less mechanical stress on the machinery. Star-Delta starter limitations and application: The starter requires additional components such as star and delta contactors and a timer, making it more expensive and requiring more maintenance than a DOL starter; it is used for medium and large motors, typically from 5 HP up to about 100 HP, and only for motors designed to run normally in delta connection. Comparison summary: DOL starters are simple, cheap, and suited to small loads but stress the supply system, while Star-Delta starters are costlier and more complex but provide reduced starting current and torque, making them the preferred choice for larger induction motors.	3. Explain the importance of Personal Protective Equipment (PPE) in electrical safety with examples. Definition: Personal Protective Equipment (PPE) refers to specialised clothing, gear, and devices worn or used by workers to protect themselves from electrical hazards such as electric shock, arc flash, burns, and falls while working on or near live electrical systems. Importance: PPE acts as the last line of defence between a worker and a hazard, especially where equipment cannot be completely de-energised or where residual risk remains. Proper use of PPE significantly reduces the severity of injuries, prevents fatalities, and ensures compliance with statutory safety regulations such as those laid down under the Factories Act and Bureau of Indian Standards. Insulated gloves: Made of rubber and rated for specific voltage levels, these protect the hands from direct contact with live conductors during testing, wiring, or maintenance work. Safety helmets: These protect the head from falling objects and accidental impacts while working at heights or in switchyards and substations. Insulated safety shoes: These have non-conductive soles that prevent current from passing through the body to the ground, which is critical when standing on potentially live or wet surfaces. Flame-resistant clothing and arc-flash suits: These protect against severe burns caused by arc flash incidents, which can generate temperatures far higher than the surface of the sun for a brief instant. Safety goggles or face shields: These protect the eyes from sparks, molten metal splashes, and UV radiation produced during switching operations or welding near electrical panels. Insulated hand tools: Screwdrivers and pliers with rubber-coated handles prevent current from travelling through hand tools into the worker's body. Safety belts and harnesses: These are essential when working on overhead lines or towers to prevent fatal falls from height. Conclusion: PPE does not eliminate electrical hazards but drastically minimises the risk of injury or death when hazards cannot be fully controlled through de-energisation, making it a mandatory and indispensable part of any electrical safety programme in industrial, commercial, and domestic settings.
4. Describe the procedure to administer CPR (Cardio-Pulmonary Resuscitation) to an unconscious person. Definition: Cardio-Pulmonary Resuscitation (CPR) is an emergency life-saving procedure performed when a person's breathing or heartbeat has stopped, commonly required after an electric shock that has caused cardiac arrest, and it aims to manually maintain blood circulation and oxygen supply to the brain and vital organs. Step 1 — Ensure scene safety: In the case of electric shock, switch off the power supply or separate the victim from the source using a non-conductive object before approaching, to avoid becoming a second casualty. Step 2 — Check responsiveness: Gently tap the victim's shoulders and shout to see if they respond; if there is no response, call for emergency medical help or ask a bystander to do so immediately. Step 3 — Check breathing: Tilt the head back slightly, lift the chin, and observe the chest for movement, listen for breath sounds, and feel for airflow for about 10 seconds. Step 4 — Begin chest compressions: Place the heel of one hand at the centre of the chest, place the other hand on top and interlock the fingers, keep arms straight, and push down hard and fast to a depth of about 5 to 6 centimetres at a rate of 100 to 120 compressions per minute, allowing full chest recoil between compressions. Step 5 — Give rescue breaths: After every 30 compressions, give 2 rescue breaths by tilting the head back, lifting the chin, pinching the nose shut, sealing your mouth over the victim's mouth, and blowing steadily for about one second while watching the chest rise. Step 6 — Continue the cycle: Repeat the cycle of 30 compressions followed by 2 rescue breaths without interruption until the person shows signs of life, an AED becomes available, or trained medical personnel arrive and take over. Conclusion: If untrained in rescue breaths, hands-only continuous chest compressions should be performed instead, and prompt, correctly performed CPR within the first few minutes greatly increases the chance of survival for a victim of electric shock.	5. What is basic electrical safety? Explain the five main electrical hazards. Definition: Basic electrical safety refers to the set of practices, precautions, and standard procedures followed to prevent injury, death, fire, or damage to property caused by electricity, in both domestic and industrial environments, through proper protective equipment, safe working procedures such as lockout-tagout, correct maintenance, and awareness of emergency response. Hazard 1 — Electric shock: This occurs when the human body becomes part of an electrical circuit and current flows through it, causing burns, muscle contraction, ventricular fibrillation, or cardiac arrest; even currents as low as 30–50 mA passing through the heart can be fatal. Hazard 2 — Arc flash: This is a sudden release of intense heat and light energy caused by a short circuit or fault through the air, often reaching temperatures far higher than the surface of the sun, and can cause severe burns, blindness, and hearing damage even without direct contact. Hazard 3 — Arc blast: This is the explosive pressure wave that accompanies an arc flash, caused by rapid expansion of superheated air and vaporised metal, which can throw workers, damage equipment, and cause serious trauma from the shockwave. Hazard 4 — Fire hazard: This arises from overheating of conductors due to overloading, poor connections, or insulation failure, which can ignite nearby combustible materials and lead to a full-scale electrical fire if not controlled early by devices like fuses and circuit breakers. Hazard 5 — Explosion hazard: This occurs particularly in hazardous atmospheres containing gas leaks, chemical vapours, or dust, where an electrical spark or arc from switching equipment can ignite the atmosphere, leading to a violent explosion, requiring flameproof or intrinsically safe equipment in such locations. Conclusion: Understanding these five hazards allows workers and designers to select appropriate protective measures such as insulation, earthing, protective devices, and PPE, significantly reducing the risk of accidents in any electrical installation.	6. Explain the different types of fire extinguishers and their applications in fire safety. Fire classification: Fires are broadly classified into Class A (ordinary combustibles like wood, paper, and cloth), Class B (flammable liquids like oil and petrol), Class C (flammable gases), Class D (combustible metals), and Class E or electrical fires involving live electrical equipment. Water-type extinguishers: These work by cooling the burning material below its ignition temperature and are effective for Class A fires; however, they must never be used on electrical fires because water is a good conductor and can cause severe electric shock to the user. Foam-type extinguishers: These form a blanket over the burning liquid, cutting off its oxygen supply, and are primarily used for Class B fires, but like water types, they are not safe for use on live electrical equipment due to their water content. Dry Chemical Powder (DCP) extinguishers: These work by interrupting the chemical chain reaction of combustion and smothering the fire, and are highly versatile, being effective on Class A, B, and C fires, as well as being safe for electrical fires because the powder is non-conductive. Carbon Dioxide (CO2) extinguishers: These displace oxygen around the fire and cool the burning surface, and are particularly suitable for electrical and flammable liquid fires because CO2 gas is non-conductive and leaves no residue that could damage sensitive electronic equipment. Clean agent extinguishers: These gaseous, halon-replacement extinguishers are used in server rooms and areas with delicate electronics, as they extinguish fire without leaving any residue behind. Application in electrical installations: DCP and CO2 extinguishers are the preferred choices near switchboards, transformers, and control panels because of their non-conductive properties, while water and foam extinguishers should be kept well away from live electrical equipment, since a wrongly chosen extinguisher can worsen an electrical fire or endanger the user's life through electric shock.
7. Explain the different types of electrical safety signs and the procedure to rescue a person from electrocution. Danger signs: These are usually red and white, indicating an immediate hazard that could cause serious injury or death, such as "Danger – High Voltage" signs placed near transformers and overhead lines. Warning signs: These are typically yellow with black text or symbols, alerting people to a potential hazard requiring caution, such as "Caution – Electrical Hazard" or "Risk of Electric Shock" signs. Mandatory signs: These are usually blue and white, instructing workers on required actions, such as "Wear Insulated Gloves" or "Lockout Before Servicing." Prohibition signs: These are shown as a red circle with a diagonal line on a white background, indicating actions that must not be performed, such as "Do Not Switch On" or "No Entry – Authorised Personnel Only." Information signs: These are generally green, indicating safe conditions or the location of safety equipment such as first-aid boxes or emergency exits. Rescue Step 1 — Switch off supply: The most critical step is to switch off the power supply immediately at the nearest switch, isolator, or main distribution board. Rescue Step 2 — Separate the victim: If the supply cannot be switched off quickly, separate the victim from the live source using a non-conductive material such as a dry wooden stick, dry rope, or rubber gloves, while standing on a dry insulated surface, never using bare hands or wet/metallic objects. Rescue Step 3 — Check and respond: Once free, check the victim's responsiveness and breathing; if unconscious and not breathing normally, begin CPR immediately while someone calls for emergency medical assistance. Rescue Step 4 — First aid for burns: Cover any burns with a clean, dry cloth without applying ointment, keep the victim warm and lying down, and wait for medical help while remaining calm throughout the rescue.	8. Define voltage, current and resistance. Mention their SI units and state Ohm's Law. Voltage: Voltage, also called electromotive force or potential difference, is the amount of work done or energy required to move a unit positive charge from one point to another in a circuit, representing the "electrical pressure" that drives current through a conductor. Its SI unit is the volt (V), defined as the potential difference between two points when one joule of work is done in moving one coulomb of charge, and it is measured using a voltmeter connected in parallel across the component. Current: Current is the rate of flow of electric charge through a conductor, caused by the movement of free electrons under an applied voltage, flowing conventionally from the positive terminal to the negative terminal in the external circuit. Its SI unit is the ampere (A), defined as the flow of one coulomb of charge per second past a given point, and it is measured using an ammeter connected in series with the circuit. Resistance: Resistance is the property of a material or component that opposes the flow of current, converting electrical energy into heat, and depends on the material's resistivity, length, cross-sectional area, and temperature. Its SI unit is the ohm (Ω), defined as the resistance that allows a current of one ampere to flow when one volt is applied across it, and it is measured using an ohmmeter or multimeter. Ohm's Law statement: Ohm's Law states that, at constant temperature, the current flowing through a conductor is directly proportional to the potential difference applied across its two ends, provided all physical conditions remain unchanged. Formula and application: This is expressed as $V = I \times R$, where V is voltage in volts, I is current in amperes, and R is resistance in ohms; from this, $I = V/R$ and $R = V/I$. Ohm's Law forms the fundamental basis for analysing and designing electrical circuits, applying to ohmic conductors like metallic resistors, though it does not hold for non-linear devices such as diodes and transistors.	9. Explain the behaviour of current and voltage in a parallel circuit with examples. Definition: A parallel circuit is one in which two or more components are connected across the same pair of nodes, so each component has its own independent path for current, and all components share the same two end points. Voltage behaviour: In a parallel circuit, the voltage across each branch is always the same and equal to the supply voltage, because all branches are connected directly across the same source terminals; this is the defining feature that distinguishes parallel circuits from series circuits, where voltage divides. Current behaviour: The total current drawn from the source equals the sum of the individual currents flowing through each parallel branch, as required by Kirchhoff's Current Law, meaning a branch with lower resistance draws more current than one with higher resistance, even though voltage across every branch is identical. Example setup: Consider a 12 V supply connected across three resistors in parallel: $R_1 = 4\ \Omega$, $R_2 = 6\ \Omega$, and $R_3 = 12\ \Omega$. Step 1 — Branch currents: Since voltage across each is 12 V, using Ohm's Law: $I_1 = 12/4 = 3\text{ A}$, $I_2 = 12/6 = 2\text{ A}$, and $I_3 = 12/12 = 1\text{ A}$. Step 2 — Total current: $I_T = I_1 + I_2 + I_3 = 3 + 2 + 1 = 6\text{ A}$. Step 3 — Equivalent resistance: $1/R_{eq} = 1/4 + 1/6 + 1/12 = 3/12 + 2/12 + 1/12 = 6/12$, so $R_{eq} = 2\ \Omega$; checking, $I_T = V/R_{eq} = 12/2 = 6\text{ A}$, which matches. Practical relevance: This behaviour explains why household wiring uses parallel connections for appliances and lights — each device receives the full supply voltage of 230 V independently, and if one device fails or is switched off, the others continue operating normally since they are on separate paths, unlike a series circuit where a single break would stop current to all devices.

Fundamentals of Electrical and Electronics Engineering

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10. State Ohm's Law and solve a numerical on resistors connected in series (equivalent resistance, current and voltage drop). Ohm's Law: Ohm's Law states that, at a constant temperature, the current flowing through a conductor is directly proportional to the potential difference applied across its ends, expressed as $V = I \times R$, where V is voltage in volts, I is current in amperes, and R is resistance in ohms. Series circuit behaviour: In a series circuit, resistors are connected end to end so the same current flows through each resistor, while the total supply voltage divides among them in proportion to their resistance values; the equivalent resistance is the sum of individual resistances: $R_{eq} = R_1 + R_2 + R_3$. Given data: Three resistors, $R_1 = 5\ \Omega$, $R_2 = 10\ \Omega$, and $R_3 = 15\ \Omega$, are connected in series across a 60 V DC supply; find the equivalent resistance, circuit current, and voltage drop across each resistor. Step 1 — Equivalent resistance: $R_{eq} = R_1 + R_2 + R_3 = 5 + 10 + 15 = 30\ \Omega$. Step 2 — Circuit current: Since the same current flows through all series resistors, $I = V/R_{eq} = 60/30 = 2\text{ A}$, which flows through R_1, R_2, and R_3 alike. Step 3 — Voltage drop across each resistor: Using Ohm's Law individually: $V_1 = I \times R_1 = 2 \times 5 = 10\text{ V}$; $V_2 = I \times R_2 = 2 \times 10 = 20\text{ V}$; $V_3 = I \times R_3 = 2 \times 15 = 30\text{ V}$. Verification: The sum of the individual voltage drops should equal the supply voltage, as required by Kirchhoff's Voltage Law: $V_1 + V_2 + V_3 = 10 + 20 + 30 = 60\text{ V}$, which matches the given supply voltage, confirming the calculations are correct, and illustrating that current stays constant while larger resistors drop proportionally more voltage in a series circuit.	11. Explain the sinusoidal AC waveform. Define amplitude, time period, frequency, cycle and phase difference with their units. Definition of AC waveform: An alternating current (AC) waveform is a periodic waveform in which the magnitude and direction of voltage or current vary continuously with time, and a sinusoidal AC waveform is one that follows the mathematical shape of a sine function, expressed as $v(t) = V_m \sin(\omega t)$, where V_m is the maximum value and ω is the angular frequency. Generation of sinusoidal waveform: This waveform is generated when a coil rotates at a uniform angular velocity within a uniform magnetic field, producing an EMF that varies sinusoidally as the angle between the coil and the field changes, which is the principle used in AC generators (alternators). Amplitude: Amplitude, also called peak value or maximum value, is the highest value reached by the voltage or current waveform in either the positive or negative direction, measured from the zero reference line. Its unit is the volt (V) for voltage waveforms and the ampere (A) for current waveforms. Time period: Time period is the time taken by the waveform to complete one full cycle of variation, that is, the time after which the waveform repeats itself exactly. Its SI unit is the second (s), and it is usually denoted by T. Frequency: Frequency is the number of complete cycles the waveform completes in one second, and is the reciprocal of time period, expressed as $f = 1/T$. Its SI unit is the hertz (Hz); for example, the standard supply frequency in India is 50 Hz. Cycle: A cycle refers to one complete set of positive and negative variations of an alternating quantity, that is, the waveform starting from zero, rising to positive maximum, returning to zero, falling to negative maximum, and returning to zero again. It has no unit since it represents one complete repetition of the waveform pattern. Phase difference: Phase difference is the angular displacement between two alternating quantities of the same frequency that do not reach their corresponding maximum or zero values at the same instant, indicating how much one waveform leads or lags another. It is measured in degrees ($^\circ$) or radians (rad), and is denoted by the symbol ϕ (phi). Importance: Understanding these parameters is essential for analysing AC circuits, designing power systems, and studying the behaviour of AC machines, since amplitude determines voltage/current ratings, frequency determines the speed of rotating machines, and phase difference determines power factor and the interaction between multiple AC sources or components such as resistors, inductors, and capacitors in a circuit.	12. Explain electrical work, power, energy and power factor along with their SI units and the meters used to measure them. Electrical work: Electrical work is the work done in moving electric charge through a circuit due to an applied potential difference, and it is calculated as $W = V \times Q$, where V is voltage and Q is charge; in practical circuits it is also expressed as $W = V \times I \times t$. Its SI unit is the joule (J), and it does not have a dedicated separate meter, as it is derived from power and time measurements. Electrical power: Electrical power is the rate at which electrical work is done or electrical energy is consumed or generated, expressed as $P = V \times I$ for DC circuits, and as $P = V \times I \times \cos\phi$ for AC circuits where $\cos\phi$ is the power factor. Its SI unit is the watt (W), and it is measured using a wattmeter, which has both a current coil and a voltage (pressure) coil to give the true power reading. Electrical energy: Electrical energy is the total electrical work done over a period of time, expressed as $E = P \times t$, and represents the actual consumption of electricity by a load. Its SI unit is the joule (J), but for practical and commercial purposes it is measured in kilowatt-hour (kWh), commonly called a "unit" of electricity. It is measured using an energy meter (kWh meter), which may be an electromechanical induction-disc type or a modern digital/smart meter installed at consumer premises. Power factor: Power factor is defined as the cosine of the phase angle between voltage and current in an AC circuit ($\cos\phi$), and it indicates how effectively the electrical power is being converted into useful work output; it ranges from 0 to 1, with a power factor of 1 (unity) indicating purely resistive load and maximum efficiency. It has no unit since it is a dimensionless ratio, and it is measured using a power factor meter, or calculated using readings from a wattmeter, voltmeter, and ammeter together. Practical significance: A low power factor increases the current drawn for the same real power, causing higher I^2R losses, larger conductor sizing requirements, and penalties from electricity boards, making power factor correction using capacitor banks an important practice in industrial installations.
13. Explain the function of RCCB and ELCB in electrical protection. RCCB definition: A Residual Current Circuit Breaker (RCCB) is a protective device that continuously monitors the balance of current flowing between the live (phase) and neutral conductors of a circuit and automatically disconnects the supply when it detects a leakage or imbalance beyond a preset threshold, typically 30 mA for human protection. RCCB working principle: Under normal healthy conditions, the current flowing into a circuit through the live conductor is exactly equal to the current returning through the neutral conductor, so their net magnetic effect on a built-in current transformer (core balance transformer) inside the RCCB is zero. If a fault occurs, such as current leaking to earth through a person's body or damaged insulation, an imbalance is created between the live and neutral currents; this imbalance induces a small voltage in the sensing coil, which trips a relay mechanism and opens the contacts, disconnecting the supply within milliseconds. RCCB application: RCCBs primarily protect human life against electric shock caused by earth leakage or indirect contact with live parts, and they are widely installed in domestic and commercial distribution boards to protect against fatal shocks from faulty appliances or damaged wiring, though they do not protect against overload or short-circuit currents on their own. ELCB definition: An Earth Leakage Circuit Breaker (ELCB) is an older type of protective device that also detects earth leakage current, but unlike the RCCB, the voltage-operated type of ELCB senses a rise in potential on the earth conductor itself relative to the general mass of earth, rather than sensing a current imbalance between live and neutral. ELCB working principle: A voltage-operated ELCB is connected between the earth wire of the installation and a separate earth electrode; if a fault causes the earthed metal casing of an appliance to rise to a dangerous voltage, this voltage difference is sensed by the ELCB's relay coil, which trips the breaker and disconnects the supply. Comparison and application: RCCBs are more reliable and are now the industry-preferred standard because they directly measure current imbalance regardless of how the fault occurs, whereas voltage-operated ELCBs depend on a proper independent earth electrode and are increasingly being phased out; both devices, however, serve the common purpose of protecting against electric shock and fire caused by earth faults in electrical installations.	14. Describe the necessity of protective devices and explain any four protective devices with suitable diagrams. Necessity of protective devices: Protective devices are essential in electrical installations to safeguard human life, equipment, and property against the harmful effects of faults such as overloads, short circuits, earth leakage, and over-voltage conditions. Without protective devices, excessive current flowing during a fault could cause overheating of conductors, insulation damage, fire hazards, and electric shocks to users, while sensitive equipment could be permanently damaged. Protective devices detect abnormal conditions and automatically isolate the faulty section of the circuit, minimising damage and ensuring continuity of supply to the healthy parts of the system. Fuse: A fuse is a simple protective device consisting of a thin wire or metal strip with a low melting point, connected in series with the circuit; when the current exceeds its rated value, the fuse element heats up and melts, breaking the circuit and disconnecting the supply. The diagram shows the fuse wire mounted in a fuse carrier, connected in series between the supply and the load; fuses are cheap and simple but must be manually replaced after operation. Miniature Circuit Breaker (MCB): An MCB is an automatic switching device that protects circuits against overload and short-circuit conditions using a combination of a thermal element (bimetallic strip) for overload protection and an electromagnetic element (solenoid) for short-circuit protection; when a fault occurs, the tripping mechanism releases the contacts, opening the circuit. Unlike a fuse, an MCB can be reset simply by switching it back on after the fault is cleared. RCCB: As explained earlier, this device senses current imbalance between live and neutral conductors and trips the circuit to protect against earth leakage and electric shock, providing protection that fuses and MCBs cannot offer since they do not sense earth faults involving small leakage currents. Earthing (protective earthing): Earthing connects the metallic body of electrical equipment to the general mass of earth through a low-resistance conductor, so that in case of insulation failure, fault current flows safely to earth instead of through a person touching the equipment, and it also helps operate protective devices like fuses and MCBs quickly by providing a low-resistance fault path. Together, these four devices — fuse, MCB, RCCB, and earthing — provide layered protection against different types of electrical faults.	15. Draw and explain the wiring diagram for controlling one lamp from two different places. Purpose: This wiring arrangement, commonly known as two-way switching, allows a single lamp to be switched ON or OFF independently from two different locations, such as controlling a staircase light from both the ground floor and the first floor, or a corridor light from either end. Components used: The circuit uses two two-way switches (each having one common terminal and two travelling terminals), one lamp, and connecting wires from the supply, without needing any additional intermediate switch for just two control points. Circuit diagram description: The phase wire from the supply is connected to the common (COM) terminal of the first two-way switch. The two travelling terminals of the first switch are connected via two wires to the two travelling terminals of the second two-way switch. The common (COM) terminal of the second switch is then connected to one terminal of the lamp, while the other terminal of the lamp is connected to the neutral wire of the supply. Working explanation: When both switches are in positions such that the common terminal of switch 1 connects to travelling wire 1, and the common terminal of switch 2 also connects to the same travelling wire 1, the circuit is complete and the lamp glows. If a person operates either switch, changing its internal connection from travelling wire 1 to travelling wire 2, the circuit path is broken at that switch, and the lamp turns OFF, because now the two commons are connected through different travelling wires. Operating the switch again at either location restores continuity through the alternate travelling wire, so the lamp turns back ON. In this way, the lamp can be turned on or off from either switch location regardless of the position of the other switch, since the completion of the circuit depends only on both switches aligning to the same travelling wire, which can be achieved by changing the state of either switch. This principle is extended using intermediate switches when control from more than two locations is required.
16. Explain the different types of earthing (pipe earthing and plate earthing) with neat diagrams. Purpose of earthing: Earthing, or grounding, is the process of connecting the non-current-carrying metallic parts of electrical equipment, or the neutral point of a system, to the general mass of earth through a conductor of negligible resistance, in order to provide a safe path for fault current and protect both equipment and human life from electric shock. Pipe earthing: In pipe earthing, a galvanised iron or steel pipe, typically 38 to 50 mm in diameter and about 2 to 3 metres long, is vertically buried in the ground at a moist location, with small holes drilled along its lower portion to allow better contact with the surrounding soil. The pipe is surrounded by alternate layers of charcoal and salt to maintain low soil resistance and keep the earth moist, improving conductivity. An earth wire is connected from the equipment to the top of the pipe through a galvanised iron clamp, and a funnel arrangement is often provided at the top for periodically pouring water to maintain moisture, especially during dry seasons. Pipe earthing is widely used because it is relatively simple to install, requires less space, and provides reliable low-resistance earthing suitable for most residential and industrial installations. Plate earthing: In plate earthing, a copper plate (typically 60 cm × 60 cm × 3.18 mm) or a galvanised iron plate (60 cm × 60 cm × 6.3 mm) is buried vertically in the ground at a depth of at least 3 metres below ground level, surrounded by alternate layers of charcoal and salt similar to pipe earthing, to reduce earth resistance. An earth wire is securely bolted to the plate and brought up through a GI pipe to the surface for connection to the installation, and a watering pipe is also provided to keep the surrounding soil moist. Plate earthing generally provides lower and more stable earth resistance than pipe earthing but requires more excavation, space, and cost due to the greater depth and larger plate size involved. Comparison: Pipe earthing is more economical, easier to install, and commonly used for smaller installations, while plate earthing offers better and more consistent performance but at higher installation cost, making it more suitable for larger industrial or high-current installations where lower earth resistance is critical.	17. Describe the different types of wiring systems (surface conduit, concealed conduit, PVC casing-capping) with their applications. Purpose of wiring systems: Electrical wiring systems refer to the methods used to run and protect cables within a building, ensuring safety, durability, and ease of maintenance while distributing power to various points such as lights, sockets, and appliances. Surface conduit wiring: In this system, PVC or metallic conduit pipes are mounted on the surface of walls or ceilings using clamps or saddles, and the electrical wires are drawn through these conduits, which are then connected to switch boxes, junction boxes, and distribution boards. This system is easy to install, allows easy identification of faults, and permits easy addition or alteration of wiring at a later stage since the conduit is visible and accessible. It is commonly used in industrial buildings, workshops, godowns, and temporary installations where appearance is less important than ease of maintenance and modification. Concealed conduit wiring: In this system, conduit pipes are embedded within the walls, floors, or ceilings before plastering, so that only the switch boards, sockets, and light points are visible on the finished surface, while all the wiring remains hidden inside the conduit within the structure. This gives a neat, aesthetically pleasing appearance, protects the wiring from mechanical damage and dust, and is widely used in residential buildings, offices, hotels, and modern commercial complexes; however, it is more expensive to install, and fault-finding or alteration after construction is more difficult since it requires breaking open the wall. PVC casing-capping wiring: In this system, wires are laid inside a PVC casing (a rectangular channel), which is fixed to the wall or ceiling, and then covered with a matching PVC capping that snaps or screws onto the casing to enclose and protect the wires. This method is cheaper than conduit wiring, reasonably easy to install and maintain, and provides some mechanical protection to the wires, but it is less durable, more prone to damage from moisture, rodents, and physical impact, and is generally used in low-budget residential wiring and temporary structures rather than in high-rise or industrial buildings. Application summary: Surface conduit suits industrial and temporary setups needing easy access, concealed conduit suits permanent residential and commercial buildings needing a clean finish, and casing-capping suits economical, low-load domestic wiring where cost is a primary concern.	18. Describe the working principle of a single-phase transformer and explain its transformation ratio. Definition: A single-phase transformer is a static electrical machine that transfers electrical energy from one circuit to another at the same frequency, but usually at a different voltage level, through the principle of mutual electromagnetic induction, without any moving parts. Construction: It consists of a laminated silicon-steel core, which provides a low-reluctance path for magnetic flux, and two windings — the primary winding, connected to the AC supply, and the secondary winding, connected to the load — both wound on the same core but electrically insulated from each other. Working principle: When an alternating voltage is applied to the primary winding, it causes an alternating current to flow through it, which sets up an alternating magnetic flux in the transformer core, in accordance with Faraday's law of electromagnetic induction. This alternating flux links with both the primary and secondary windings as it flows through the common core. Because the flux is continuously changing, it induces an EMF in the primary winding (which opposes the applied voltage, called back-EMF, limiting the current drawn) and also induces an EMF in the secondary winding, which is connected to the external load. If the secondary circuit is closed through a load, this induced EMF drives a current through the load, thereby transferring electrical energy from the primary side to the secondary side purely through magnetic coupling, without any direct electrical connection between the two windings. Transformation ratio: The transformation ratio (or turns ratio) of a transformer is defined as the ratio of the number of turns in the secondary winding to the number of turns in the primary winding, and it determines whether the transformer steps up or steps down the voltage. It is expressed as $K = N_2/N_1 = V_2/V_1 = I_1/I_2$, where N_1 and N_2 are the number of turns, V_1 and V_2 are the primary and secondary voltages, and I_1 and I_2 are the primary and secondary currents. If K is greater than 1, the transformer is a step-up transformer, increasing voltage while decreasing current; if K is less than 1, it is a step-down transformer, decreasing voltage while increasing current, always maintaining the same power (VA) on both sides in an ideal transformer.

19. Explain the difference between single-phase and three-phase induction motors. Single-phase induction motor definition: A single-phase induction motor operates on a single-phase AC supply and is generally used for low-power applications, typically up to about 1 HP, because a single-phase supply alone produces a pulsating (non-rotating) magnetic field that cannot produce starting torque on its own. Starting mechanism: Since a single alternating flux cannot create a rotating magnetic field by itself, single-phase motors require additional starting arrangements such as a starting winding with a capacitor, a shaded pole arrangement, or a centrifugal switch to create a temporary phase difference and produce enough starting torque to begin rotation; once running, the motor continues to rotate due to the interaction of the main winding flux with the induced rotor currents. Three-phase induction motor definition: A three-phase induction motor operates on a three-phase AC supply, where the three phase windings are displaced 120 electrical degrees apart in space and are supplied with currents that are 120 degrees apart in time, which automatically produces a naturally rotating magnetic field in the stator without needing any extra starting device. Self-starting nature: Because the three-phase supply itself creates a rotating magnetic field, three-phase induction motors are inherently self-starting, and the rotating field induces currents in the rotor conductors (by electromagnetic induction), which then interact with the stator field to produce continuous torque and rotation, following the basic induction motor principle. Comparison of features: Single-phase motors have lower efficiency, lower power factor, lower starting torque, and are used for small loads such as fans, mixers, pumps, and small workshop tools, whereas three-phase motors have higher efficiency, higher power factor, higher starting torque, and are used for large industrial loads such as pumps, compressors, conveyors, lathes, and heavy machinery. Construction difference: Single-phase motors typically have two stator windings (main and auxiliary/starting winding), while three-phase motors have three identical stator windings connected in star or delta configuration; both types use either squirrel-cage or wound rotors, though squirrel-cage rotors are more common in both. Application summary: Single-phase motors are preferred in domestic and light commercial applications due to availability of single-phase supply, whereas three-phase motors are preferred in industrial applications where higher power, better efficiency, smoother torque, and continuous operation are required.	20. Explain the different types of batteries and their selection criteria for various applications. Definition: A battery is a device that stores electrical energy in chemical form and converts it back into electrical energy through electrochemical reactions when connected to a load, and batteries are broadly classified into primary (non-rechargeable) and secondary (rechargeable) types. Primary batteries: These batteries can be used only once, as the chemical reaction inside them is irreversible; examples include zinc-carbon batteries and alkaline batteries, which are commonly used in low-drain devices like remote controls, torches, and clocks due to their low cost and long shelf life, but they cannot be recharged once exhausted. Lead-acid battery: This is a secondary (rechargeable) battery using lead dioxide and sponge lead electrodes with dilute sulphuric acid as the electrolyte; it is widely used in automobiles for starting, lighting, and ignition (SLI), as well as in UPS systems and inverters, due to its high surge current capability, low cost, and reliability, though it is heavy and has a relatively shorter cycle life compared to newer technologies. Nickel-Cadmium (Ni-Cd) battery: This rechargeable battery offers good performance at low temperatures and a long cycle life, and is used in power tools, emergency lighting, and aviation equipment, though it suffers from the "memory effect" and contains toxic cadmium, raising environmental concerns. Lithium-ion (Li-ion) battery: This modern rechargeable battery offers high energy density, lightweight construction, low self-discharge, and no memory effect, making it the preferred choice for laptops, mobile phones, electric vehicles, and portable electronic devices, though it requires a protection circuit and is more expensive than lead-acid batteries. Selection criteria: The selection of a suitable battery for a given application depends on several factors: the required voltage and current (capacity) rating, the energy density and weight constraints of the application, the expected cycle life and how frequently the battery will be charged and discharged, the operating temperature range, the cost and budget available, the charging time required, safety considerations, and environmental/disposal concerns. For example, high energy density and light weight are prioritised for portable electronics and electric vehicles, whereas high surge current and low cost are prioritised for automotive starting batteries, and long cycle life and reliability are prioritised for UPS and backup power applications.	21. Explain the concept of Ampere-Hour capacity and efficiency of a battery. Ampere-Hour capacity definition: The Ampere-Hour (Ah) capacity of a battery is a measure of the amount of electric charge the battery can deliver at a specified discharge rate over a specified period of time before its voltage drops to a defined cut-off level, and it is calculated as the product of the discharge current in amperes and the discharge time in hours, expressed as $Ah = I \times t$. Significance of Ah rating: For example, a battery rated at 100 Ah can theoretically deliver a current of 100 A for 1 hour, or 10 A for 10 hours, or 5 A for 20 hours, though in practice the actual deliverable capacity decreases somewhat at higher discharge rates due to internal losses, a phenomenon described by Peukert's law. This rating helps engineers determine how long a battery can supply a given load before requiring recharge, and is essential for sizing batteries in UPS systems, inverters, electric vehicles, and solar power backup systems. Factors affecting Ah capacity: The actual available capacity of a battery depends on factors such as the discharge rate (higher discharge rates reduce effective capacity), ambient temperature (capacity generally decreases at low temperatures), the age and state of health of the battery, and the depth of discharge to which it is subjected during use. Battery efficiency definition: Battery efficiency, also called coulombic or energy efficiency, is the ratio of the electrical energy (or charge) delivered by the battery during discharge to the electrical energy (or charge) supplied to it during charging, expressed as a percentage: $\text{Efficiency} = (\text{Energy output during discharge} / \text{Energy input during charging}) \times 100$. Significance of efficiency: A battery is never 100 percent efficient because some energy is lost as heat due to internal resistance, and some is lost in side chemical reactions during charging and discharging; typical lead-acid batteries have an efficiency of around 70 to 85 percent, while lithium-ion batteries typically achieve higher efficiencies of around 90 to 95 percent. Practical importance: Understanding both Ah capacity and efficiency is essential for correctly sizing battery banks for a given load, estimating backup time accurately, determining the charging current and time required, and comparing the performance of different battery technologies for a specific application such as UPS backup, solar energy storage, or electric vehicles.
22. Explain the types, selection criteria and sizing procedure of a UPS for a given load. Definition: A UPS (Uninterruptible Power Supply) is a device that provides continuous, backup electrical power to connected loads during a mains power failure, using a battery bank and an inverter circuit, ensuring that critical equipment such as computers, servers, and medical devices do not suffer sudden shutdowns. Offline (standby) UPS: In this type, the load is normally supplied directly from the mains supply, and the UPS switches to battery-inverter operation only when the mains fails, with a small switchover time (typically a few milliseconds); it is the simplest and cheapest type but is suitable only for non-critical loads that can tolerate a brief interruption, such as personal computers. Line-interactive UPS: This type continuously monitors and regulates the incoming mains voltage using an autotransformer, correcting minor voltage fluctuations without switching to battery, and switches to battery power only during a complete outage or major voltage deviation, offering better protection than offline UPS at a moderate cost, commonly used for small offices and networking equipment. Online (double-conversion) UPS: In this type, the load is continuously supplied through the inverter, which is continuously powered by the battery, which in turn is continuously charged by the rectified mains supply, so there is zero switchover time and complete isolation from mains disturbances; this type provides the highest quality, most reliable power and is used for critical loads such as data centres, hospitals, and industrial control systems, though it is the most expensive and least energy-efficient type due to continuous conversion losses. Selection criteria: The choice of UPS type depends on the criticality of the load (whether even a few milliseconds of interruption is acceptable), the required power quality (voltage stability, protection from spikes/sags), the budget available, and the acceptable running cost/efficiency. Sizing procedure: First, calculate the total connected load in VA or watts by summing the power ratings of all equipment to be connected, and apply a suitable safety margin (typically 20–25%) for future expansion and inrush currents. Second, determine the required backup time in hours based on how long power must be maintained during an outage. Third, calculate the required battery Ah capacity using the formula $Ah = (\text{Load in watts} \times \text{Backup time in hours}) / (\text{Battery voltage} \times \text{Inverter efficiency})$, and select standard battery capacities that meet or exceed this value. Finally, select a UPS with a VA/kVA rating higher than the calculated load, and choose an appropriate battery bank voltage and configuration (series/parallel) to match the UPS's DC input requirement.	23. Explain the difference between full-wave and half-wave rectifiers with suitable diagrams. Half-wave rectifier definition: A half-wave rectifier uses only a single diode connected in series with the AC supply and the load, and it converts only one half cycle of the AC input into DC output, blocking the other half cycle completely. Half-wave working: During the positive half cycle of the AC input, the diode is forward biased and conducts, allowing current to flow through the load, producing an output pulse. During the negative half cycle, the diode becomes reverse biased and blocks current flow completely, so the output voltage remains zero during this half. The diagram shows the AC source connected in series with a single diode and the load resistor, with output taken across the load. Half-wave characteristics: The output is highly pulsating with large gaps between pulses, the ripple frequency equals the supply frequency (50 Hz), the average DC output voltage is low, and the rectification efficiency is low (about 40.6%), making it unsuitable for most practical applications without heavy filtering. Full-wave rectifier definition: A full-wave rectifier converts both half cycles of the AC input into DC output, and can be implemented either using a centre-tapped transformer with two diodes, or using a bridge rectifier with four diodes (as explained earlier), producing output current during both halves of the input cycle. Full-wave working: In the centre-tapped version, one diode conducts during the positive half cycle and the other conducts during the negative half cycle, but current flows through the load in the same direction in both cases, giving continuous output pulses without gaps between them. Full-wave characteristics: The output has a higher average DC voltage, the ripple frequency is twice the supply frequency (100 Hz), which makes filtering easier and more effective, and the rectification efficiency is higher (about 81.2%) compared to the half-wave rectifier. Comparison summary: Full-wave rectifiers (bridge or centre-tapped) are far more efficient, provide smoother DC with less ripple, and better utilize the transformer, whereas half-wave rectifiers are simpler and cheaper but produce a lower-quality, highly pulsating DC output suitable only for low-power, non-critical applications.	24. Compare conductors, insulators and semiconductors with examples. Conductors: Conductors are materials that allow electric current to flow through them easily because they have a large number of free electrons in their outermost (valence) shell that can move freely under the influence of an applied electric field. They have very low resistivity (typically in the order of 10^{-8} ohm-metre), no forbidden energy gap between the valence band and conduction band (the two bands overlap), and their conductivity generally decreases slightly with an increase in temperature due to increased atomic vibrations. Common examples include copper, aluminium, silver, and gold, which are widely used for electrical wiring, cables, and conductive components due to their excellent conductivity. Insulators: Insulators are materials that strongly oppose the flow of electric current because their valence electrons are tightly bound to the atoms and are not free to move, resulting in very high resistivity (typically in the order of 10^{10} to 10^{16} ohm-metre or higher). Insulators have a very wide forbidden energy gap (greater than about 5 eV) between the valence band and conduction band, preventing electrons from easily jumping to the conduction band. Common examples include rubber, glass, mica, PVC, and porcelain, which are used to cover conductors, support overhead lines, and prevent unwanted current leakage or short circuits. Semiconductors: Semiconductors are materials whose conductivity lies between that of conductors and insulators, and whose electrical properties can be significantly altered by adding controlled impurities (doping), applying heat, or applying light. They have a narrow forbidden energy gap (typically around 0.7 to 1.1 eV, for example 0.7 eV for germanium and 1.1 eV for silicon) between the valence and conduction bands, small enough that some electrons can cross it at room temperature, and unlike conductors, their conductivity increases with an increase in temperature. Common examples include silicon and germanium, which form the basis of diodes, transistors, integrated circuits, and other electronic devices. Comparison summary: Conductors have overlapping bands and very low resistivity suited for carrying current, insulators have a wide energy gap and very high resistivity suited for preventing current flow, and semiconductors have a small controllable energy gap that allows their conductivity to be precisely engineered, making them the foundation of modern electronic devices such as diodes, transistors, and integrated circuits.
25. Explain the operation of a relay as an actuator in a control circuit. Definition: A relay is an electromechanical (or solid-state) switching device that uses a low-power control signal to control a much larger current or voltage in a separate circuit, acting as an actuator that translates an electrical control signal into a mechanical switching action. Construction: An electromechanical relay consists of an electromagnetic coil, an iron core, a movable armature, a spring, and one or more sets of contacts (normally open and/or normally closed), all mounted within a housing, with the coil circuit (control side) electrically isolated from the contact circuit (load side). Working principle: When a control voltage is applied to the relay's coil terminals, current flows through the coil, creating an electromagnetic field around the iron core. This magnetic field attracts the movable armature, pulling it towards the core against the tension of the return spring. As the armature moves, it mechanically operates the attached contacts, closing the normally open (NO) contacts and/or opening the normally closed (NC) contacts, thereby completing or breaking the circuit connected to the load side. When the control voltage is removed, the electromagnetic field collapses, and the spring pulls the armature back to its original (de-energised) position, restoring the contacts to their normal state. Role as an actuator: In a control circuit, a relay acts as an actuator because it converts a small electrical control signal (which might come from a sensor, microcontroller, PLC, or low-power switch) into a mechanical action that switches a completely separate, often much higher-power circuit, such as a motor, heater, lamp, or industrial machine. This allows low-voltage, low-current control systems (such as microcontroller output pins that cannot handle high current directly) to safely and reliably control high-voltage, high-current loads without direct electrical connection between the control and load circuits, since the coil and contacts are electrically isolated. Applications: Relays are widely used in automatic control panels, motor starters, home automation systems, automobile electrical systems, and industrial automation, wherever a low-power signal needs to switch a higher-power device safely, reliably, and with electrical isolation between control and power circuits.	26. Explain the structure, symbol and working principle of a PN junction diode and describe its use as a switch. Structure: A PN junction diode is formed by joining a P-type semiconductor (doped with trivalent impurities, creating majority charge carriers called holes) with an N-type semiconductor (doped with pentavalent impurities, creating majority charge carriers called free electrons) within a single semiconductor crystal, typically silicon or germanium. The diode has two terminals: the anode, connected to the P-side, and the cathode, connected to the N-side, usually marked by a band on the physical device. Symbol: The circuit symbol of a PN junction diode is a triangle pointing towards a straight line, where the triangle represents the anode (P-side) and the line represents the cathode (N-side); current flows conventionally in the direction the triangle points, from anode to cathode, when forward biased. Formation of depletion region: At the moment the P and N regions are joined, free electrons from the N-side diffuse across the junction into the P-side, and holes from the P-side diffuse into the N-side, leaving behind a region near the junction depleted of free charge carriers, called the depletion region, which creates a small built-in potential barrier (about 0.7 V for silicon and 0.3 V for germanium) opposing further diffusion. Forward bias working: When the positive terminal of an external supply is connected to the P-side (anode) and the negative terminal to the N-side (cathode), the applied voltage opposes and narrows the depletion region; once the applied voltage exceeds the barrier potential, the depletion region collapses, and the diode conducts current freely with very low resistance. Reverse bias working: When the polarity is reversed (positive to N-side, negative to P-side), the depletion region widens further, increasing the barrier, so only a negligible reverse saturation (leakage) current flows, and the diode effectively blocks current, behaving almost like an open circuit. Use as a switch: A PN junction diode functions as a simple electronic switch because it behaves like a closed switch (very low resistance, conducting) when forward biased above its threshold voltage, and like an open switch (very high resistance, non-conducting) when reverse biased, allowing it to control the direction and flow of current in circuits such as rectifiers, clipping/clamping circuits, and logic gate implementations without any moving parts, offering fast switching speed and long operational life.	27. Explain the structure and working principle of a BJT (transistor) and describe how it works as an electronic switch. Structure: A Bipolar Junction Transistor (BJT) is a three-terminal semiconductor device formed by sandwiching one type of semiconductor material between two layers of the opposite type, creating two PN junctions; it exists in two configurations — NPN (a P-type layer sandwiched between two N-type layers) and PNP (an N-type layer sandwiched between two P-type layers). The three terminals are the Emitter (E), which is heavily doped and supplies the majority charge carriers; the Base (B), which is very thin and lightly doped, controlling the flow of carriers; and the Collector (C), which is moderately doped and collects the carriers that pass through the base. Working principle (NPN transistor): For normal (active mode) operation, the emitter-base junction is forward biased, and the collector-base junction is reverse biased. The forward bias at the emitter-base junction causes a large number of electrons to be injected from the emitter into the thin base region; because the base is very thin and lightly doped, only a small fraction of these electrons recombine with holes in the base (forming the small base current), while the vast majority are swept across into the collector region due to the reverse-biased collector-base junction's electric field, forming the much larger collector current. This gives the fundamental relationship $I_C = \beta \times I_B$, where β (current gain) is typically large (50–300), meaning a small base current controls a much larger collector current — the basis of transistor amplification. Operation as a switch: A transistor operates as an electronic switch by being driven between two extreme operating regions — cut-off and saturation — rather than the linear active region used for amplification. In the cut-off region, no base current is applied (or the base-emitter junction is not sufficiently forward biased), so no collector current flows, and the transistor behaves like an open switch between collector and emitter. In the saturation region, a sufficiently large base current is applied, driving the transistor into full conduction, where it offers minimum resistance between collector and emitter, behaving like a closed switch, allowing maximum collector current to flow (limited mainly by the external load resistance). Practical application: By applying a digital control signal (0 or logic-high voltage) to the base through a suitable base resistor, the transistor can be switched rapidly between cut-off (OFF) and saturation (ON) states, making it the fundamental building block for digital logic circuits, relay driver circuits, motor control circuits, LED drivers, and power switching applications in microcontroller and PLC-based automation systems.

